

# SPRINT 11 – Recruitment API Integration using Queueable Apex

Introduction: This sprint focused on integrating a Salesforce Application record with an external Recruitment API. The solution uses Queueable Apex for asynchronous HTTP callouts, a Named Credential for the external endpoint, External Credential access controls, and Application fields to store the integration result.

Key concepts: External API Integration, HTTP Callout, Queueable Apex, Database.AllowsCallouts, Named Credential, External Credential Principal, Permission Set, JSON Serialization/Deserialization, HTTP 201 response, Integration Status and Integration Error.

Application fields: Application Number, Applied Date, Student, Job, Status, Integration Status, External Candidate Id, Last Integration Attempt and Integration Error.

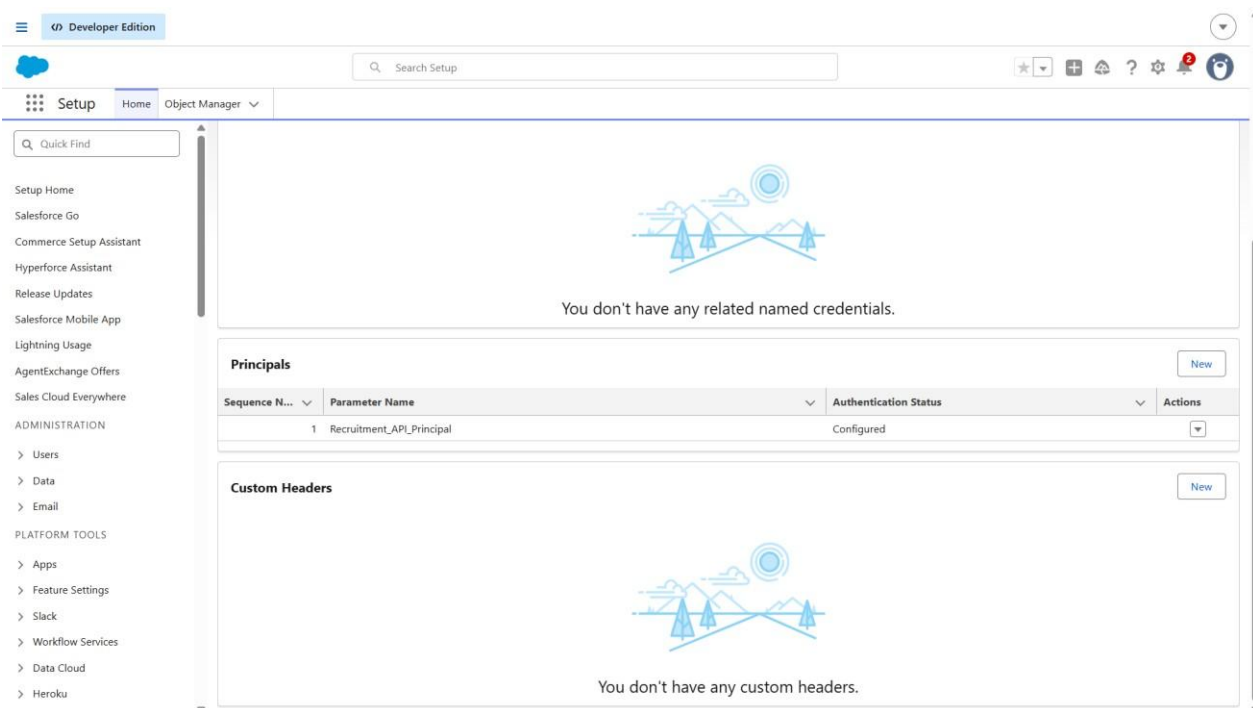
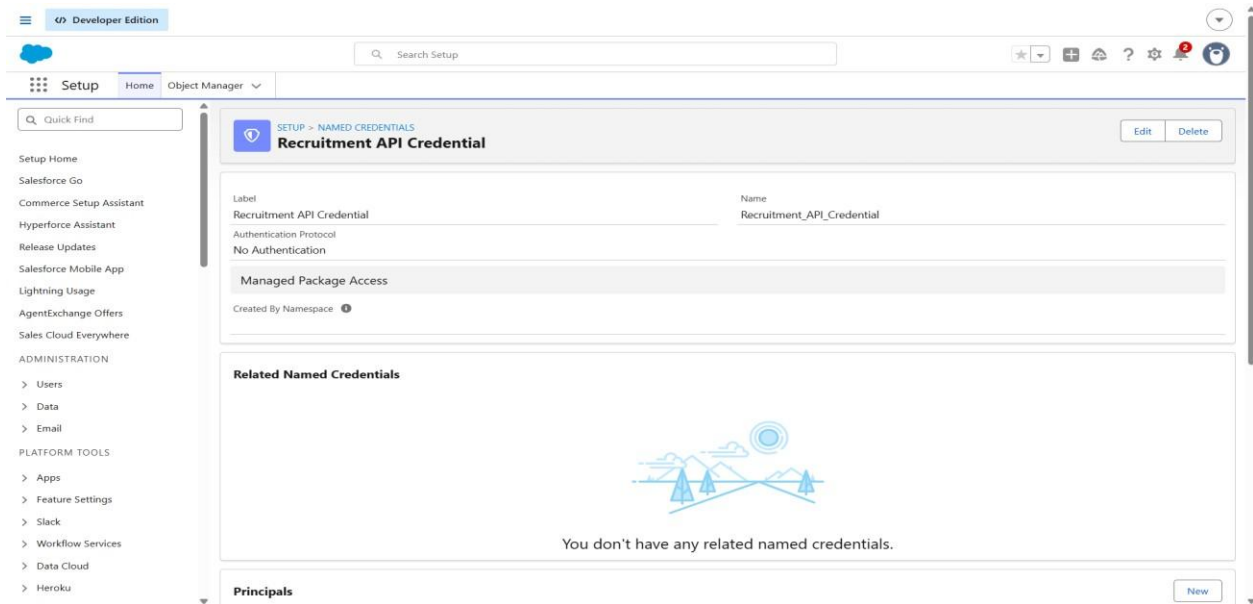
|                                                            |                             |                       |                   |         |
|------------------------------------------------------------|-----------------------------|-----------------------|-------------------|---------|
| SETUP > OBJECT MANAGER                                     |                             |                       |                   |         |
| Application                                                |                             |                       |                   |         |
| Details                                                    |                             |                       |                   |         |
| Fields & Relationships                                     |                             |                       |                   |         |
| 12 Items, Sorted by Field Label                            |                             |                       |                   |         |
| Q Quick Find                                               |                             |                       |                   |         |
| New Deleted Fields Field Dependencies Set History Tracking |                             |                       |                   |         |
| FIELD LABEL                                                | FIELD NAME                  | DATA TYPE             | CONTROLLING FIELD | INDEXED |
| Application Number                                         | Name                        | Text(80)              |                   | ✓       |
| Applied Date                                               | Applied_Date__c             | Date                  |                   |         |
| Created By                                                 | CreatedById                 | Lookup(User)          |                   |         |
| External Candidate Id                                      | External_Candidate_Id__c    | Text(100)             |                   |         |
| Integration Error                                          | Integration_Error__c        | Long Text Area(32768) |                   |         |
| Integration Status                                         | Integration_Status__c       | Picklist              |                   |         |
| Job                                                        | Job__c                      | Lookup(Job)           |                   | ✓       |
| Last Integration Attempt                                   | Last_Integration_Attempt__c | Date/Time             |                   |         |
| Last Modified By                                           | LastModifiedById            | Lookup(User)          |                   |         |
| Owner                                                      | OwnerId                     | Lookup(User,Group)    |                   | ✓       |
| Status                                                     | Status__c                   | Picklist              |                   |         |

## Named Credentials

A Named Credential named “Recruitment API Credential” was configured for the external recruitment service. Apex refers to the configured callout name instead of hard-coding the endpoint or authentication secrets.

Queueable Apex Implementation: CandidateSyncQueueable implements Queueable and Database.AllowsCallouts. It receives the Application Id, prepares candidate data, serializes the data to JSON, sends the request and reads the response.

Architecture: Application → Queueable Apex → HTTP Request → Named Credential → Recruitment API.



## Queueable Apex – Request Construction

**Class and request setup:** the Queueable class performs the background integration and configures the HTTP request with the Named Credential endpoint, POST method and JSON content type.

**Candidate data and JSON body:** candidate details are collected from the Application and related records, placed in a Map and converted into JSON using `JSON.serialize`.

**Design benefit:** the integration logic is isolated from the trigger and LWC, making the solution easier to test, maintain and troubleshoot.

```

1 public class CandidateSyncQueueable implements Queueable, Database.AllowsCallouts {
2
3     private Id applicationId;
4
5     public CandidateSyncQueueable(Id applicationId) {
6         this.applicationId = applicationId;
7     }
8
9     public void execute(QueueableContext context) {
10
11         HttpRequest request = new HttpRequest();
12
13         request.setEndpoint(
14             'callout:Recruitment_API/posts'
15         );
16
17         request.setMethod('POST');
18
19         request.setHeader(
20             'Content-Type',
21             'application/json'
22         );
23
24         Map<String, Object> candidateData =
25             new Map<String, Object>();
26
27         candidateData.put('applicationId', applicationId);
28         candidateData.put('studentId', 'STU1001');
29         candidateData.put('name', 'Pravallika');
30         candidateData.put('email', 'pravallika@example.com');
31         candidateData.put('branch', 'AIIML');
32
33     }
34 }

```

```

27         candidateData.put('applicationId', applicationId);
28         candidateData.put('studentId', 'STU1001');
29         candidateData.put('name', 'Pravallika');
30         candidateData.put('email', 'pravallika@example.com');
31         candidateData.put('branch', 'AIIML');
32         candidateData.put('cgpa', 8.5);
33         candidateData.put('jobId', 'JOB1001');
34         candidateData.put('company', 'ABC Technologies');
35         candidateData.put('role', 'Salesforce Developer');
36
37         request.setBody(
38             JSON.serialize(candidateData)
39         );
40
41         Http http = new Http();
42
43         try {
44
45             HttpResponse response =
46                 http.send(request);
47
48             Integer statusCode =
49                 response.getStatusCode();
50
51             System.debug(
52                 'Response Status: ' + statusCode
53             );
54
55             System.debug(
56                 'Response Body: ' +
57                 response.getBody()
58             );
59         } catch (Exception e) {
60             System.debug('Error: ' + e.getMessage());
61         }
62     }
63 }

```

## Queueable Apex – Code and Payload

The request setup, candidate mapping and JSON construction are kept inside CandidateSyncQueueable. The business process initiates the job, while the Queueable class owns the external communication.

The request body contains the candidate information required by the Recruitment API. JSON serialization creates a structured payload without manually concatenating strings.

This separation improves reusability, testing and maintainability.

```

37  request.setBody(
38      JSON.serialize(candidateData)
39  );
40
41  Http http = new Http();
42
43  try {
44
45      HttpResponse response =
46          http.send(request);
47
48      Integer statusCode =
49          response.getStatusCode();
50
51      System.debug(
52          'Response Status: ' + statusCode
53      );
54
55      System.debug(
56          'Response Body: ' +
57          response.getBody()
58      );
59
60  } catch (Exception e) {
61
62      System.debug(
63          'Callout Error: ' +
64          e.getMessage()
65      );
66  }
67  }
68  }

```

## Response Handling

**Successful response:** when the API returns HTTP 201, the response body is deserialized, Integration Status becomes Sent, the returned external id is stored and Integration Error is cleared.

Failure response: for a non-success status code, Integration Status becomes Failed and the API status and response body are stored as the error.

Exception handling: unexpected exceptions are caught, the failure is recorded and Last Integration Attempt is updated.

The Salesforce record therefore reflects the actual result of the external operation.

The screenshot displays the Salesforce Developer Edition interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The left sidebar shows 'Users' and 'Permission Sets'. The main content area is titled 'SETUP Permission Sets' and shows the 'Recruitment API Access' permission set. Below this, there is a section for 'External Credential Principal Access' with a table listing the 'Recruitment\_API\_Credential - Recruitment\_API\_Principal' as an 'Installed Package'.

## Permission Set and External Credential Access

The Recruitment API Access permission set provides access to the external credential principal. This keeps integration access controlled through Salesforce security configuration.

The integration should not expose credentials in Apex source code. Access is granted deliberately through Salesforce security mechanisms.

The result is a controlled connection between the Queueable integration and the configured external credential.

Developer Edition

Setup

Home

Object Manager

per

Hyperforce Assistant

Users

Permission Set Groups

Permission Sets

Apps

Mobile Apps

Salesforce

Mobile Builder for the Seller-Focused Experience

Feature Settings

Digital Experiences

Salesforce CMS

Settings

Functions

Sales

Accounts

Person Accounts

Process Automation

Automation App

Flows

Migrate to Flow

Setup

PERMISSION SET 'RECRUITMENT API ACCESS'

Recruitment API Access

Current Assignments

Full Name

Active

Role

Profile

User License

Expires On

PADALA DURGA SSHA PRAV...

✓

System Administrator

Salesforce

Add Assignment

File Edit Debug Test Workspace Help

CandidateSyncQueueable.apex

Log executeAnonymous @8/11/2026, 11:51:43 AM

Log executeAnonymous @8/11/2026, 12:12:04 PM

Log executeAnonymous @8/11/2026, 12:12:05 PM

Execution Log

| Timestamp    | Event      | Details                            |
|--------------|------------|------------------------------------|
| 12:12:05:612 | USER_DEBUG | [51]DEBUG:Response Status: 201     |
| 12:12:05:612 | USER_DEBUG | [55]DEBUG:Response Body: {         |
| 12:12:05:000 | USER_DEBUG | "role": "Salesforce Developer",    |
| 12:12:05:000 | USER_DEBUG | "company": "ABC Technologies",     |
| 12:12:05:000 | USER_DEBUG | "jobId": "JOB1001",                |
| 12:12:05:000 | USER_DEBUG | "cpgs": 8.5,                       |
| 12:12:05:000 | USER_DEBUG | "branch": "AIML",                  |
| 12:12:05:000 | USER_DEBUG | "email": "pravallika@example.com", |
| 12:12:05:000 | USER_DEBUG | "name": "Pravallika",              |
| 12:12:05:000 | USER_DEBUG | "studentId": "STU1001",            |
| 12:12:05:000 | USER_DEBUG | "applicationId": null,             |
| 12:12:05:000 | USER_DEBUG | "id": 101                          |
| 12:12:05:000 | USER_DEBUG | }                                  |

Loading...

This Frame

Executable

Debug Only

Filter: USER\_DEBUG

Logs

Tests

Checkpoints

Query Editor

View State

Progress

Problems

| User                         | Application | Operation                                | Time                   | Status  | Read   | Size      |
|------------------------------|-------------|------------------------------------------|------------------------|---------|--------|-----------|
| PADALA DURGA SSHA PRAVALLIKA | Unknown     | QueueableHandler                         | 8/11/2026, 12:12:05 PM | Success |        | 8.03 KB   |
| PADALA DURGA SSHA PRAVALLIKA | Unknown     | /services/data/v67.0/tooling/executeA... | 8/11/2026, 12:12:04 PM | Success |        | 3.85 KB   |
| PADALA DURGA SSHA PRAVALLIKA | Browser     | /udd/PermissionSet/viewPermissionSet...  | 8/11/2026, 12:09:53 PM | Success | Unread | 527 bytes |
| PADALA DURGA SSHA PRAVALLIKA | Browser     | /udd/PermissionSet/editPermissionSet...  | 8/11/2026, 12:06:32 PM | Success | Unread | 527 bytes |
| PADALA DURGA SSHA PRAVALLIKA | Browser     | /udd/PermissionSet/viewPermissionSet...  | 8/11/2026, 12:05:20 PM | Success | Unread | 527 bytes |
| PADALA DURGA SSHA PRAVALLIKA | Unknown     | Batch Apex                               | 8/11/2026, 12:00:03 PM | Success | Unread | 3.68 KB   |
| PADALA DURGA SSHA PRAVALLIKA | Unknown     | SerialBatchApexRangeChunkHandler         | 8/11/2026, 12:00:03 PM | Success | Unread | 5.1 KB    |

Filter Click here to filter the log list

# Testing and Execution

The Queueable Apex job was executed from the Developer Console. The execution logs showed successful QueueableHandler processing and a successful HTTP response.

Observed response: HTTP status 201 with external id 101, indicating that the candidate submission was successfully accepted or created by the external service.

Developer Console logs were used to verify the response status and response body.

Developer Edition

Q Search...

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👤

Playground Starter

Welcome

Install a Package

Get Your Login Credentials

\* APP-005 | Application

✕

✎

👤

Application

APP-005

New Contact

Edit

New Opportunity

▼

Related

Details

Application Number

APP-005

✎

Student

[SOWJANYA Updated](#)

✎

Job

[INFOSYS](#)

✎

Status

Applied

✎

Applied Date

11/08/2026

✎

Integration Status

Pending

✎

External Candidate Id

✎

Last Integration Attempt

✎

Integration Error

✎

Owner

👤

[PADALA DURGA SESHA PRAVALLIKA](#)

🔔

Created By

👤

[PADALA DURGA SESHA PRAVALLIKA](#)

11/08/2026, 12:17 pm

Last Modified By

👤

[PADALA DURGA SESHA PRAVALLIKA](#)

11/08/2026, 12:17 pm

## Successful Integration Result



File • Edit • Debug • Test • Workspace • Help • < >

Log executeAnonymous @8/11/2026, 12:20:33 PM • Log executeAnonymous @8/11/2026, 12:20:34 PM • ApplicationTrigger.appt • Log executeAnonymous @8/11/2026, 2:24:18 PM • Log executeAnonymous @8/11/2026, 2:27:03 PM

### Execution Log

| Timestamp    | Event      | Details                                                                                                                   |
|--------------|------------|---------------------------------------------------------------------------------------------------------------------------|
| 14:27:03:000 | USER_INFO  | [EXTERNAL]005hg000001eh2U(23pa1ak281@curious-raccoon-5fzpx8.com)[GMT+05:30] India Standard Time (Asia/Kolkata)[GMT+05:30] |
| 14:27:03:983 | USER_DEBUG | [84]DEBUG Response Status: 201                                                                                            |
| 14:27:03:983 | USER_DEBUG | [88]DEBUG Response Body: {                                                                                                |
| 14:27:03:000 | USER_DEBUG | "appliedDate": "2026-08-11",                                                                                              |
| 14:27:03:000 | USER_DEBUG | "jobId": "a01hg000006p2QvAAT",                                                                                            |
| 14:27:03:000 | USER_DEBUG | "studentId": "a00hg00000F1Q3hAA3",                                                                                        |
| 14:27:03:000 | USER_DEBUG | "applicationNumber": "APP-005",                                                                                           |
| 14:27:03:000 | USER_DEBUG | "applicationId": "a02hg000001JW744AG",                                                                                    |
| 14:27:03:000 | USER_DEBUG | "id": 101                                                                                                                 |
| 14:27:03:000 | USER_DEBUG | }                                                                                                                         |

Loading...

This Frame Executable Debug Only Filter USER

| User                          | Application | Operation        | Time                  | Status  | Read   | Size     |
|-------------------------------|-------------|------------------|-----------------------|---------|--------|----------|
| PADALA DURGA SESHA PRAVALLIKA | Browser     | /aura            | 8/11/2026, 2:27:03 PM | Success | Unread | 5.51 KB  |
| PADALA DURGA SESHA PRAVALLIKA | Unknown     | QueueableHandler | 8/11/2026, 2:27:03 PM | Success |        | 13.12 KB |
| PADALA DURGA SESHA PRAVALLIKA | Browser     | /aura            | 8/11/2026, 2:26:26 PM | Success | Unread | 3.93 KB  |
| PADALA DURGA SESHA PRAVALLIKA | Browser     | /aura            | 8/11/2026, 2:24:18 PM | Success |        | 3.92 KB  |

Filter Click here to filter the log list

Developer Edition

Q Search...

Playground Starter Welcome Install a Package Get Your Login Credentials APP-005 | Application X

Application APP-005

New Contact Edit New Opportunity

Related Details

|                          |                                                     |                  |                                                    |
|--------------------------|-----------------------------------------------------|------------------|----------------------------------------------------|
| Application Number       | APP-005                                             | Owner            | PADALA DURGA SESHA PRAVALLIKA                      |
| Student                  | SOWIANYA Updated                                    |                  |                                                    |
| Job                      | INFOSYS                                             |                  |                                                    |
| Status                   | Selected                                            |                  |                                                    |
| Applied Date             | 11/08/2026                                          |                  |                                                    |
| Integration Status       | Sent                                                |                  |                                                    |
| External Candidate Id    | 101                                                 |                  |                                                    |
| Last Integration Attempt | 11/08/2026, 2:27 pm                                 |                  |                                                    |
| Integration Error        |                                                     |                  |                                                    |
| Created By               | PADALA DURGA SESHA PRAVALLIKA, 11/08/2026, 12:17 pm | Last Modified By | PADALA DURGA SESHA PRAVALLIKA, 11/08/2026, 2:30 pm |

## Overall Integration Workflow

1. An Application record is created or updated in Salesforce.
2. Application information is passed to the asynchronous integration process.
3. CandidateSyncQueueable executes in the background and is allowed to make callouts.
4. Apex creates an HTTP POST request through the configured Named Credential.
5. Candidate information is stored in a Map and serialized into JSON.
6. Salesforce sends the JSON payload to the external Recruitment API.



7. The API returns a response; successful runs return HTTP 201 and external id 101.
8. Salesforce stores the returned external candidate id and changes Integration Status to Sent.
9. Last Integration Attempt records the synchronization time.

The screenshot displays the Salesforce Developer Console. The top pane shows the 'Execution Log' for a recent anonymous Apex execution. The log contains several 'USER\_DEBUG' statements that output the response body of an API call, which includes fields like 'appliedDate', 'studentId', 'applicationNumber', 'applicationId', and 'id'.

The bottom pane shows a table of application records. The table has columns for User, Application, Operation, Time, Status, Read, and Size. The records show successful API calls and their corresponding application status updates.

| User                          | Application | Operation        | Time                  | Status  | Read   | Size     |
|-------------------------------|-------------|------------------|-----------------------|---------|--------|----------|
| PADALA DURGA SETHA PRAVALLIKA | Browser     | /aura            | 8/11/2026, 2:50:44 PM | Success | Unread | 3.93 KB  |
| PADALA DURGA SETHA PRAVALLIKA | Browser     | /aura            | 8/11/2026, 2:49:24 PM | Success | Unread | 5.54 KB  |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | QueueableHandler | 8/11/2026, 2:49:24 PM | Success | Unread | 13.67 KB |
| PADALA DURGA SETHA PRAVALLIKA | Browser     | /aura            | 8/11/2026, 2:37:59 PM | Success | Unread | 3.93 KB  |
| PADALA DURGA SETHA PRAVALLIKA | Browser     | /aura            | 8/11/2026, 2:30:47 PM | Success | Unread | 3.93 KB  |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | QueueableHandler | 8/11/2026, 2:27:03 PM | Success | Unread | 13.12 KB |

## Problems Encountered and Resolution

- The integration was refined so that the Application record records the result of the API operation instead of only sending data.
- Success handling: set Integration Status to Sent, store the returned external id and clear the error.
- Failure handling: set Integration Status to Failed and save the API response as an integration error.
- Exception handling: capture the exception message and update the Application record.
- Monitoring: use Developer Console debug logs to verify response status and response body.